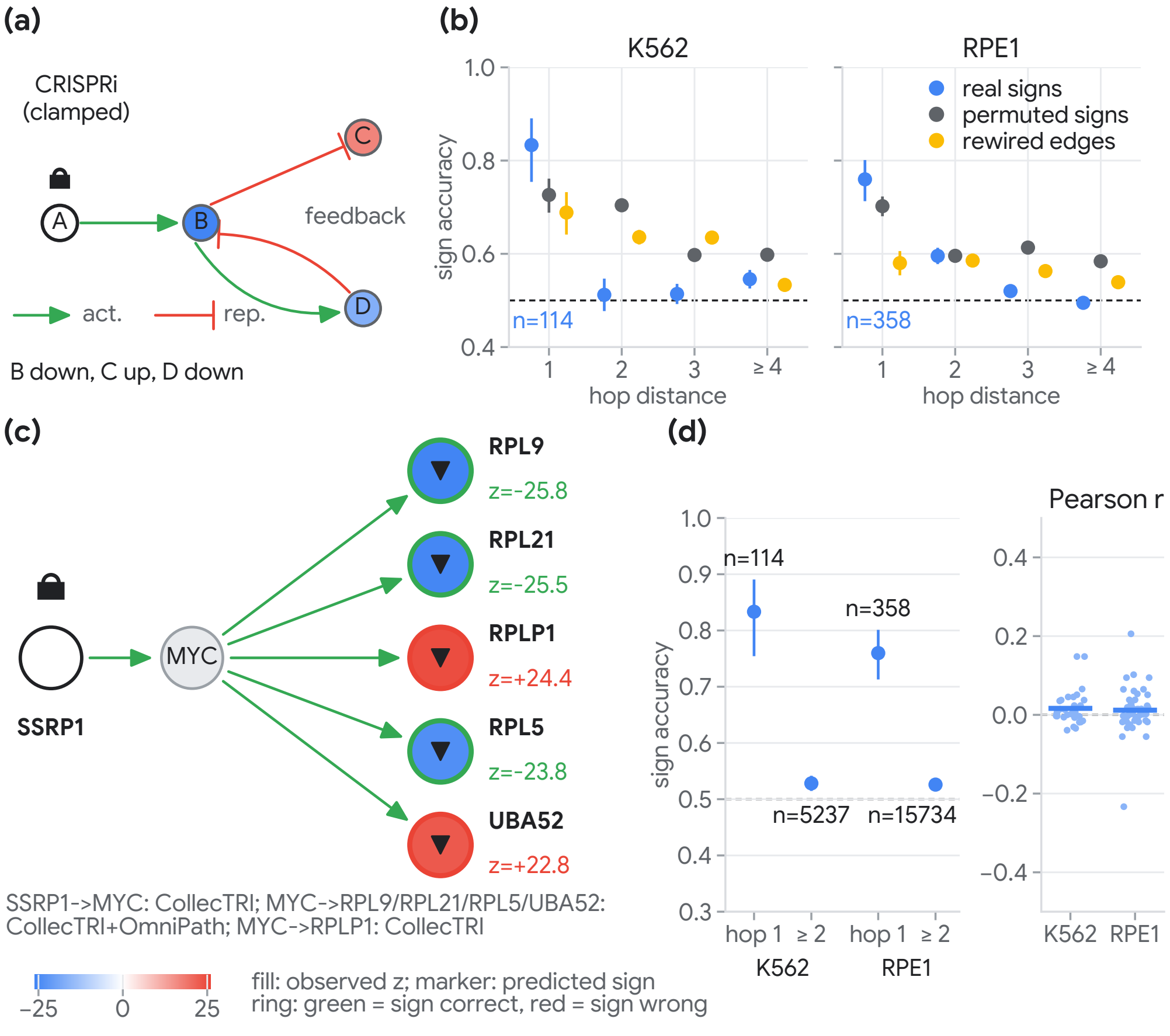
(a)
CRISPRi
(clamped)
A
B
C
D
feedback
act.
rep.
B down, C up, D down
(b)
K562
RPE1
real signs
permuted signs
rewired edges
sign accuracy
1.0
0.8
0.6
0.4
n=114
n=358
1
2
3
≥ 4
hop distance
(c)
SSRP1
MYC
RPL9
z=-25.8
RPL21
z=-25.5
RPLP1
z=+24.4
RPL5
z=-23.8
UBA52
z=+22.8
SSRP1->MYC: CollecTRI; MYC->RPL9/RPL21/RPL5/UBA52:
CollecTRI+OmniPath; MYC->RPLP1: CollecTRI
−25
0
25
fill: observed z; marker: predicted sign
ring: green = sign correct, red = sign wrong
(d)
sign accuracy
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
n=114
n=358
n=5237
n=15734
hop 1
≥ 2
K562
RPE1
Pearson r
0.4
0.2
0.0
−0.2
−0.4
K562
RPE1